

OCNA: mDNS and NAT (Which IP Address Is Visible)

Study notes for the Omada OCNA exam.

mDNS (Multicast DNS)

- Lets devices find each other on a local network **without a DNS server**. A device asks everyone at once, and the device that owns the name answers.
- **Port:** UDP 5353. **Multicast address:** 224.0.0.251 (IPv4), ff02::fb (IPv6). **Domain:** .local
- Used by Bonjour/Avahi, AirPlay, AirPrint, Chromecast and printers. DNS-SD is used to discover services.
- **Important:** mDNS is link-local multicast, so it **does not cross a router**. It only works within one VLAN/subnet.
- Typical problem: an iPhone in VLAN 10 cannot see an AirPlay device or printer in VLAN 20. Fix: put the devices in the same VLAN, or enable an **mDNS reflector/relay** on the gateway (in Omada this is the gateway's mDNS feature; the exact name and location depend on the controller version).
- A similar protocol is LLMNR (UDP 5355, Windows). The exam may ask you to tell them apart.

Which IP Address Is Visible When Communicating Through NAT

Situation	What the other side sees
Two devices on the same LAN	Each other's private addresses (192.168.x.x)
Two devices, each behind a different router , communicating over the internet	The public (WAN) IP of the other side's router , not the private address
Site-to-site VPN between two networks	Each other's private IPs (NAT is not used inside the tunnel; subnets must not overlap)
Double NAT / CGNAT (ISP)	The ISP's public IP (or an address from 100.64.0.0/10 on the inner layer)

Example

Device A has 192.168.1.10 behind a router with WAN address 203.0.113.5. Device B has 192.168.2.20 behind a router with WAN address 198.51.100.7. A connects to B via 198.51.100.7 (port forwarding must be configured there). B sees the source as **203.0.113.5**, not 192.168.1.10. The reply goes back to that public address, and A's router forwards it to A using its NAT table.

Rules to Remember

- With NAT you always see the public address of the other side's router. Private addresses are never sent across the internet.
- Only the outgoing direction can start a connection (the router creates a mapping, PAT/NAPT). Incoming connections from the internet need **port forwarding**.
- If the question mentions a **VPN tunnel**, the answer is the private IP.